CSL100/MAL100

Tutorial 5

Function

Question 1.

Write a function Name Digit_length(int a). Take any integer value from user and passed that value to function. Function calculate the length of Integer (Number of digit in given input) and return the length. And display the length

i/p: 123456
o/p: 6
i/p: 7810
o/p: 4

```c
#include <stdio.h>
// Function to calculate the length of an integer (number of digits)
int Digit_length(int a) {
        int length = 0;

        // If the number is zero, it has one digit
        if (a == 0) {
        return 1;
        }

        // Make the number positive if it's negative
        if (a < 0) {
        a = -a;
        }

        // Count the number of digits
        while (a != 0) {
        a /= 10;
        length++;
        }

        return length;
}
```

```c
int main() {
        int num;

        // Taking input from user
        printf("Enter an integer: ");
        scanf("%d", &num);

        // Calling the function and
displaying the result
        int length = Digit_length(num);
        printf("The number of digits in %d
is: %d\n", num, length);

        return 0;
}
```

Question 2:

Write a function name Palindrome(int a). Take input integer value from user and passed to the function. And function will calculate the given value is palindrome or not. If given number is Palindrome then function return value 1 otherwise it will return 0.

i/p: 11011
o/p: 1
i/p: 1234231
o/p: 0

```c
#include <stdio.h>
// Function to check if a number is a palindrome
int Palindrome(int a) {
        int original = a, reversed = 0, remainder;

        // Make the number positive if it's negative
        if (a < 0) {
        return 0;
        }
        while (a != 0) {
        remainder = a % 10;
        reversed = reversed * 10 + remainder;
        a /= 10;
        }

        if (original == reversed) {
        return 1; // The number is a palindrome
        } else {
        return 0; // The number is not a palindrome
        }
}
```

```c
int main() {
        int num;
        int re;
        // Taking input from user
        printf("Enter an integer: ");
        scanf("%d", &num);

        // Calling the function and displaying the
result
        re = palindrome(num);
        printf(“%d”,re);
        return 0;
}
```

Question 3

Write a function named Perfect_number(int a). Take input from user upper and lower range value and passed all the value from range one by one to function. The function will calculate given value is perfect number or not. Function return 1 if given number is perfect_number otherwise it will return 0. Then print all the perfect number for user given range

(a perfect number is a positive integer that is equal to the sum of its positive divisors, excluding the number itself. For instance, 6 has divisors 1, 2 and 3 (excluding itself), and 1 + 2 + 3 = 6, so 6 is a perfect number.)

Example:
I/P : lower bound = 1

Upper bound = 100
O/P: The perfect numbers between 1 to 100 are: 6 28

```c
#include <stdio.h>
int Perfect_number(int a) {
        int sum = 0;
        for (int i = 1; i <= a / 2; i++) {
        if (a % i == 0) {
        sum += i;
        }
        }

        if (sum == a && a != 0) {
        return 1; // It is a perfect number
        } else {
        return 0; // Not a perfect number
        }
}
```

```c
int main() {
        int lower, upper;
        printf("Enter the lower bound: ");
        scanf("%d", &lower);
        printf("Enter the upper bound: ");
        scanf("%d", &upper);
        printf("The perfect numbers between %d and %d are: ",
lower, upper);
        int found = 0;  // To track if any perfect number is found
        for (int i = lower; i <= upper; i++) {
        if (Perfect_number(i)) {
        printf("%d ", i);
        found = 1;
        }
        }
        if (!found) {
        printf("None");
        }

        printf("\n");

        return 0;
}
```

Question 4

Write a function to perform the following: take digit decimal number from the user as input. Remove all zeros from the number, and return the value and then Display the final result.

Example:

I/P: 105020
O/Pt: 152

```c
#include <stdio.h>
int Remove_zeros(int a) {
        int result = 0, multiplier = 1;

        // Loop through the digits of the number
        while (a != 0) {
        int digit = a % 10;
        a /= 10;

        // If the digit is not zero, add it to the result
        if (digit != 0) {
        result += digit * multiplier;
        multiplier *= 10;
        }
        }

        return result;
}
```

```c
int main() {
        int num;

        // Taking input from user
        printf("Enter a decimal number: ");
        scanf("%d", &num);

        // Calling the function and displaying the result
        int result = Remove_zeros(num);
        printf("The number after removing all zeros is:
%d\n", result);

        return 0;
}
```

QUIZ

1. Write a C program using the function to check whether a given number is a magic number or not. Pass the input to function Magic number(int a). Inside that calculate the sum of its digits then call the function Reverse(int a) to cal culate the reverse of the sum of the digits and return to the function. and then calculate the product of the sum of digits and its reverse and check whether that number is a magic number or not. if it is a magic number return 1 otherwise return 0.
   Example:
   I/P : Enter the number N = 1729
   output :- Magic Number
   1729 is a magic number.
   (Since, 1+7+2+9=19 and 19*91=1729.) 10 marks

2. Write a user-defined function that takes a number as input from the user and prints whether the given number is prime or not 5 Marks